



Investigation report

Rejected take-off and runway overrun, Tarlton Springs - NTSB-style
probable cause report

Drafted 30/09/2026

INVESTIGATION WORKFLOW SUITE

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Executive summary

What happened

On 18/06/2026, at about 17:30 local time (07:30 UTC), Saltbush Air SB512, VH-SBQ, a 50-seat twin-turboprop, was taking off from runway 09 at Tarlton Springs Aerodrome in darkness with 44 passengers, 2 flight crew and 1 cabin crew. It was the crew's fifth sector of the day and departed 40 minutes behind schedule.

At 07:30:31, 2 seconds after V1, the first officer saw two kangaroos crossing onto the runway. The pilot in command rejected the take-off at 118 kt, 6 kt above V1. One second later, the No. 2 propeller and right main gear struck both animals, and No. 2 reverse was lost. The aircraft left the sealed surface at approximately 21 kt and stopped upright on the extended centreline, 62 m into the 90 m runway end safety area and 28 m short of the ungraded terrain beyond it.

There were no serious injuries. One cabin crew member reported bruising while disembarking by the forward airstair; there was no evacuation. The aircraft was substantially damaged:

- three No. 2 propeller blades damaged beyond limits
- right main gear doors distorted
- both right main tyres deflated
- lower right fuselage punctured.

Both kangaroos were killed. The aerodrome was closed for 4 hours 20 minutes.

What the investigation found

The reject above V1 and the loss of No. 2 reverse together reduced deceleration approximately 15% below the scheduled accelerate-stop performance, which assumed a reject at or before V1 with a 67 m margin. This resulted in the overrun.

The animals very likely entered through a 140 m breach in the northern perimeter fence, recorded by the aerodrome operator in 04/2023 with animals passing through it. Its repair was deferred at three budget rounds, and it was very likely that this contributed. The applicable standard did not require an aerodrome of this size to document its wildlife measures, which likely contributed by leaving them without an owner or funded programme. This was the least firmly established contributing factor, as a documented plan would not necessarily have secured funding for the repair.

The departure was in the hour after last light, the peak kangaroo movement period in the aerodrome's own log, and this likely contributed. No runway inspection was requested. The operator required inspections only at aerodromes listed after a reported strike, which very likely meant none was required at Tarlton Springs. The evening service was approved as a schedule change without reopening a route risk assessment restricting the aerodrome to daylight operations, and this likely contributed.

The operator's rejected take-off training addressed only aircraft system failures, and this likely contributed. However, the pilot in command stated that he knew the procedure and deliberately departed from it, so the gap may bear more on the fleet than on this decision.

Maximum braking was established within 2 seconds, against a fleet mean of 3.2 seconds. This very likely kept the aircraft within the safety area.

Found not to have contributed:

- a tailwind and a No. 2 engine power loss before the reject

- crew fatigue, which the recorded response times did not support
- partially unserviceable floor proximity emergency lighting, as there was no evacuation.

Whether edge lighting intensity impaired detection of the animals could not be determined.

What can be done

Six safety issues were identified:

- an aerodrome standard leaving smaller aerodromes' wildlife measures undocumented
- a fence repair deferred for three budget cycles
- a runway inspection requirement only a strike can trigger
- a change process that did not reopen the daylight-only route risk assessment
- rejected take-off training confined to aircraft system failures
- a floor lighting defect, undetectable by the daily test, that could affect the whole fleet.

Wildlife knowledge must reach the operators who need it without waiting for a strike to trigger action.

After the occurrence, the evening Tarlton Springs service was suspended pending the investigation. No safety action had been recorded against any of the safety issues.

Factual information

History of the flight

On 18/06/2026, Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, was scheduled from Tarlton Springs Aerodrome to Sydney. It was the crew's fifth and last sector of the day, added to their duty that morning after the rostered crew reported unfit. On board were 44 passengers, 2 flight crew and 1 cabin crew. The first officer was pilot flying.

The only runway inspection that day was completed at 21:35 UTC on 17/06/2026 (07:35 local on 18/06/2026), before the first scheduled arrival, and recorded the movement area as clear. No further inspection was carried out or requested.

The doors closed at 07:16 UTC (17:16 local), 40 minutes behind schedule, in darkness. The tower was rostered to close at 18:00 local. [NOT ESTABLISHED: time of last light or end of civil twilight at Tarlton Springs on 18/06/2026]

At 07:29:10 the tower cleared SB512 for take-off on runway 09, passing the wind as "zero nine zero at six". The tower did not offer a runway inspection and the crew did not request one. No wildlife transmission was made on the tower channel in the preceding 90 minutes.

The take-off roll began at 07:30:02. The pilot in command called "Power set, both engines good." The 80 kt cross-check was completed at 07:30:15, and V1 was called at 07:30:29, at 112 kt.

At 07:30:31, at 116 kt, the first officer called "Roos - roos, roos, on the left, they're on the -". Two kangaroos were moving onto the runway from the northern side, beyond the 1,100 m point. The first officer later estimated the range at first sight as around 200 m, possibly less.

At 07:30:32, at 118 kt, 3 seconds and 6 kt after V1, the pilot in command called "Stopping. I have control." Both power levers reached flight idle within 0.6 seconds. The aircraft was 1,152 m along runway 09, with 695 m of sealed surface remaining.

At 07:30:33 the right main landing gear and the No. 2 propeller struck both animals. Maximum braking was established at 07:30:34. At 07:30:36 the pilot in command called "Reverse - number two's not coming. Max brakes, max brakes." Reverse engaged on No. 1 only.

At 07:30:47 the pilot in command called "We're not stopping on it." The aircraft left the sealed surface at 07:30:51 at approximately 21 kt and stopped at 07:30:57, 62 m into the 90 m runway end safety area (RESA), upright on the extended centreline with the nose wheel in graded gravel.

The pilot in command set the parking brake, noted no fire indications and nothing on the board, and stated "I'm not evacuating." Both engines were shut down at 07:31:26. At 07:31:34 he instructed the cabin to remain seated. At 07:31:54 he advised the tower of no injuries reported and no fire, and requested activation of the emergency plan and a vehicle. The tower activated the aerodrome emergency plan at 07:32:10 and dispatched a vehicle.

Passengers disembarked by the forward airstair with the main cabin lights on. The airstair was heard operating at 07:34:40; the occurrence notification records disembarkation at 4 minutes 8 seconds after the aircraft stopped. No evacuation was commanded, no slides were deployed, and there was no fire.

Injuries to persons

There were no serious injuries. One cabin crew member reported bruising during disembarkation. Two kangaroos were killed.

Damage to aircraft

The aircraft was substantially damaged:

- Three No. 2 propeller blades were damaged beyond repairable limits, with leading edge deformation and one blade tip separation.
- The No. 2 engine required a shock-load inspection.
- The right main gear doors were distorted.
- Both right main tyres deflated through the fuse plugs.
- The lower right fuselage skin was abraded and punctured in two places.

Personnel information

Pilot in command. 8,410 hours total, 2,140 on type, 3 years 1 month in command. One previous turboprop type and no jet ratings. Last line check 02/12/2025, satisfactory. No prior occurrence history. The crew duty record gives his last cyclic simulator session as 11/03/2026, all items satisfactory; his interview background gives his most recent recurrent simulator check as February 2026. Duty and rest in the preceding 7 days were within company limits, with no variations. He had two rest days before the duty.

First officer. 1,655 hours total, 405 on type. Last cyclic simulator session 28/04/2026, all items satisfactory. Her interview background records a line check in May 2026. She had flown two previous sectors into Tarlton Springs, both in daylight, on 09/01/2026 and 17/02/2026. She had one rest day before the duty. The duty record gives 14 months with the company; her interview background gives 14 months in role and 17 months with the organisation, having joined in April 2025.

Duty. The duty record gives sign-on at 06:30 local, with five sectors planned to end at 18:35 local, a planned duty of 12 hours 05 minutes against a maximum of 12 hours 30 minutes. The occurrence was at 17:30 local, 11 hours 00 minutes into the duty. The pilot in command's interview background records the duty as beginning at 05:10 local.

Both pilots described themselves as tired. The pilot in command said he would have declined a further sector but was not struggling. Both returned negative drug and alcohol tests after the occurrence.

Aircraft information

VH-SBQ was configured at 20,800 kg with flap 10. [NOT ESTABLISHED: aircraft type designation, maintenance status and deferred defects before the flight]

During disembarkation, the first officer noticed unlit sections of the aft cabin floor proximity strip, with the emergency lighting armed. She raised this with the engineer the next morning, and a functional check was carried out on 24/06/2026. [NOT ESTABLISHED: result of the 24/06/2026 functional check]

Meteorological information

Runway 09 was dry. The recorded headwind component was 4 to 6 kt throughout the roll, consistent with the 07:00 UTC routine observation and the tower's 090 degrees at 6 kt. No tailwind component was recorded. V1 was calculated for a 5 kt headwind component.

Aerodrome information

Runway 09 had 1,847 m of sealed surface, followed by a 90 m RESA of graded gravel. Beyond the RESA the terrain is ungraded.

The aerodrome keeps a paper wildlife sighting log in the aerodrome office. From 01/2025 to 06/2026 it held 46 entries:

- 41 recorded kangaroos on or adjacent to the movement area.
- 38 of those 41 were timed between one hour before and two hours after last light.
- 11 recorded animals on the sealed surface of runway 09/27.

The log is not reported to any operator. On 23/06/2026 the aerodrome reporting officer confirmed that no airline had ever asked for it.

The pilot in command stated that he had seen kangaroos on the strip at Tarlton Springs four or five times and had mentioned this to ground staff, but had never filed a report. The first officer stated that none of the documents she reviewed before the flight mentioned animals: the flight plan, the charts, the ERSAs and the company aerodrome brief page.

The pilot in command described the runway lighting as a corridor of light with darkness either side. He could not say whether this affected his ability to see the animals, and had been looking at the engine page after V1. The first officer described the lighting as normal, having only two daylight landings there to compare against.

[NOT ESTABLISHED: runway edge lighting intensity setting in use; condition of the perimeter fence and its repair history; applicability of a documented wildlife management plan to the aerodrome]

The aerodrome was closed for 4 hours 20 minutes after the occurrence for inspection and removal of animal remains and propeller debris.

Recorded data

The FDR, QAR and CVR were removed by the operator's engineering representative at 12:40 UTC on 18/06/2026 and read out by the operator's flight data section.

Before the reject. Both engines were at take-off torque, 98.4% and 98.5%, until 07:30:32. No exceedance, caution or warning was recorded. Fuel flow, ITT and propeller RPM were within normal scatter.

Rejected take-off.

- Maximum braking was established 2 seconds after the decision, at 2,940 psi, with anti-skid active on all four wheels. The operator's fleet mean for rejected take-offs is 3.2 seconds.
- Impact produced lateral and vertical acceleration spikes of 0.31 g right and 1.48 g.
- No. 2 propeller RPM decayed from 07:30:33, and the No. 2 reverse interlock did not make.
- Mean deceleration from 07:30:34 to 07:30:44 was 2.6 m/s², approximately 15% below the certified accelerate-stop value for the weight and runway condition, which assumes symmetric reverse.
- Deceleration fell to 1.9 m/s² in the gravel.
- Stopping distance from 118 kt was 757 m.

Tests and research

The runway and RESA survey found two carcasses, at 1,188 m and 1,221 m from the runway 09 threshold; continuous tyre marks from 1,160 m to the runway end; and a nose wheel rut ending 62 m beyond the runway end, on the extended centreline.

Animal tissue was recovered from two No. 2 propeller blade roots. Borescope inspection of both engines found no pre-existing damage and no evidence of a power anomaly before the impact.

The right main gear structure was within limits. Brake energy absorbed was calculated at 84% of the certified maximum for the weight and speed. There was no brake fire or hydraulic leak. The anti-skid system tested serviceable. The fuselage punctures were consistent with propeller debris.

Organisational and management information

The operator's procedure for requesting a runway inspection before the first departure after dark applies to aerodromes on the Appendix 8A list, under Part A 8.14. The pilot in command stated that Tarlton Springs was not on that list. [NOT ESTABLISHED: text of Part A 8.14 and Appendix 8A, including the listing criterion]

The pilot in command stated that the evening service appeared on the roster with no accompanying briefing or change, and that every stop or go scenario in his recurrent training had been an engine failure below V1. [NOT ESTABLISHED: operator's rejected take-off training syllabus; approval process for the evening service]

Additional information

After the occurrence, the Head of Flying Operations suspended the evening Tarlton Springs service pending the outcome of the investigation.

Analysis

External Influences

Documented wildlife plan required only above a size threshold the aerodrome does not meet

It was virtually certain that the applicable standard did not require Tarlton Springs to hold a documented wildlife hazard management plan. Aerodrome Standard AS-139 imposes that requirement at 10.4.2:

- 10.4.2 The operator of a certified aerodrome must prepare, implement and maintain a documented wildlife hazard management plan where the aerodrome is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year.

Tarlton Springs met neither limb. Its heaviest scheduled aeroplane was 22,900 kg, and it handled 61,400 passenger movements in 2025. Under 10.4.3, the reasonable measures required by 10.4.1 did not need to be documented.

The aerodrome held no plan, no hazard assessment and no dispersal programme. Its only documented wildlife measure was the single inspection before the first arrival, required by Aerodrome Manual 6.3. The certified aerodrome register and the aerodrome operator's compliance file confirmed this. The aerodrome was not in breach of the standard. The matter is therefore not one of compliance. The standard's own threshold left an aerodrome of this size with a wildlife programme that was undocumented, unaudited and, in practice, unfunded.

The hazard was known at the aerodrome. The sighting log for 01/2025 to 06/2026 held 46 entries. Of these, 41 recorded kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light, and 11 placed animals on the sealed surface of runway 09/27. The log was a paper register kept in the aerodrome office. It was not reported to any operator, and no airline had ever asked for it.

The physical breach was also known. Works register item WR-221, raised in 04/2023, recorded the following about the northern boundary fence between the 700 m and 1,300 m runway markers:

- The fence was displaced and strained.
- Animals had been seen passing through it.
- Replacement was recommended at an estimated \$41,000.

The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. Item WR-263, raised in 02/2026, recorded the same section as deteriorating further. On 23/06/2026 the section was found as a continuous 140 m gap, with the bottom two wires separated from the posts and an animal pad through it at the 1,050 m mark. On the evening of the occurrence, the kangaroos were first seen past the 1,100 m point, moving onto the runway from the northern side.

The case for influence is that a documented plan would have connected these two sets of knowledge to the controls that depended on them. A plan would have given the sighting log a route to the operators using the aerodrome. It would also have placed the fence on a funded programme rather than an unfunded works register. A documented plan brings an owner and an audit trail. Each of the other controls examined in this occurrence failed for want of exactly those.

The case against is substantial:

- Section 10.4.1 obliged the aerodrome operator to take reasonable measures regardless of the threshold.

- A documented plan is not the only way to repair a fence or to pass a sighting log to an airline.
- The fence deferrals were the council's budget decisions. There is no evidence that a documented plan would have been funded any better. A plan is a document; the fence required money.

The investigation accepted the influence on the balance of this evidence. The deciding consideration was not that a plan would have guaranteed repair or disclosure. It was that the absence of any documented programme removed ownership and audit from the aerodrome's wildlife measures, and the controls that failed on the evening were the ones lacking those. This conclusion rests on inference about what a documented plan would have changed, not on direct evidence of it. It is held at a lower level of confidence than the other contributing factors in this occurrence.

The threshold led to two further factors, which are argued in their own subsections:

- the deferral of the fence repair for three budget cycles
- the requirement for runway inspection only at listed aerodromes, where the listing criterion is a strike

It was likely that the AS-139 threshold, which exempted Tarlton Springs from maintaining a documented wildlife hazard management plan, contributed to the occurrence by leaving the aerodrome's knowledge of kangaroo movement and of its failed northern fence without an owner, an audit trail or a funded programme; this conclusion is inferred and less certain than the other contributing factors.

Organisational Influences

Perimeter fence repair deferred for three budget cycles

Works register item WR-221, raised in 04/2023, recorded the northern boundary fence between the 700 m and 1,300 m runway markers as failed. Four panels were displaced, the wire was strained, and animals had been observed passing through. Replacement was estimated at \$41,000. The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds, each time for review at the next cycle. Item WR-263, raised in 02/2026, recorded the same section as deteriorating further. It was virtually certain that the repair had been deferred. The deferrals are in the aerodrome operator's own register, and the site inspection on 23/06/2026 confirmed the fence's condition.

That inspection found a continuous 140 m section of fence with the bottom two wires separated from the posts. A clear animal pad ran through it at the 1,050 m mark. The kangaroos crossed from the northern side of the runway, the same side as the breach. The carcasses were recovered at 1,188 m and 1,221 m from the runway 09 threshold, further along the runway from the pad. A fence in the condition found does not exclude kangaroos, and the pad lay directly upstream of the strike position.

The hazard the fence should have controlled was documented. The aerodrome's wildlife sighting log for 01/2025 to 06/2026 held 46 entries. Of these, 41 recorded kangaroos on or adjacent to the movement area. Of those 41, 38 fell between one hour before and two hours after last light, and 11 recorded animals on the sealed surface of runway 09/27. The original 2023 register entry had itself noted animals passing through the damaged section. The fence therefore remained unrepaired while both the defect and the animal movement it allowed were on file at the aerodrome. The absence of a required wildlife hazard management plan, which left no document to give the repair priority, is addressed in the External Influences analysis.

It was very likely that the repeated deferral of the northern fence repair left open the route by which the two kangaroos reached the runway strip.

Evening service approved as a schedule change, route risk assessment not reopened

RRA-041 issue 3 (11/2022) was the operator's current route risk assessment for Tarlton Springs. It assessed runway length and accelerate-stop margin, the single runway with no lighting redundancy, the absence of an alternate within 200 nm at night, limited engineering support, and summer heat and density altitude. It did not list wildlife. It carried the operating restriction "daylight operations only pending review of night lighting and diversion fuel policy." No issue 4 exists.

CMR-2025-118 introduced an evening rotation, six days per week, from 14/07/2025. This was the first scheduled company operation into Tarlton Springs after last light. The record assessed the change as a "schedule change only", stated "No new hazard identified", and answered "No" to whether a route risk assessment review was required. The Head of Flying Operations approved it on 22/05/2025. The record does not mention the daylight-only restriction, and RRA-041 was not reopened. The Safety Department's records show no involvement beyond receipt of the approved record.

Both documents are current controlled copies. The approver has since left the company and was not interviewed, so the reasoning behind the assessment was not obtained. On that basis, it was extremely likely that the evening service was approved without the route risk assessment being reopened. The change process asked whether the operation was new in its components: the aerodrome, aircraft type, crew qualifications and engineering support. It did not ask whether the existing assessment still held for an operation that assessment had excluded.

A review of RRA-041 against a night operation would have been the point at which the aerodrome's wildlife exposure, and Tarlton Springs' absence from the Appendix 8A list of wildlife-active aerodromes, came under examination. The contrary consideration is that the restriction named only night lighting and diversion fuel. A review confined to those two items might not have reached wildlife. For that reason the connection does not rise above likely. The limitation of the Appendix 8A listing criterion, and the absence of a runway inspection request that followed from it, are addressed in the Risk Controls analysis.

It was likely that approving the evening service without reopening RRA-041 removed the opportunity to identify the wildlife hazard at Tarlton Springs before night operations began.

Risk Controls

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

The operator had a defence against wildlife on the runway at night. Operations Manual Part A 8.14.2 required the pilot in command, at an aerodrome listed in Appendix 8A, to "request a runway inspection from the aerodrome operator prior to the first departure after last light" and not to "commence the take-off roll until the result of that inspection has been passed." At every other aerodrome, 8.14.3 required "no specific action."

Appendix 8A admitted an aerodrome only where "a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months." Nine aerodromes were listed. Tarlton Springs had never been listed. Nothing in Part A asked a crew to report a sighting, and nothing admitted an aerodrome on any other evidence. It was virtually certain that the requirement was framed this way and did not apply at Tarlton Springs.

The evidence that should have armed the control existed at the aerodrome. The aerodrome's wildlife sighting log for 01/2025 to 06/2026 held 46 entries. Forty-one recorded kangaroos on or near the movement area. Thirty-eight of those fell between one hour before and two hours after last light. Eleven recorded animals on the sealed surface of runway 09/27. The log was a paper register in the aerodrome office. It was not reported to any operator, and no airline had asked for it. None of those entries was a company strike, so none could put the aerodrome on the list.

The trigger was therefore circular. The inspection could only be required after the kind of event it existed to prevent. The control was sound. Its admission criterion left it unarmed at an aerodrome with a documented pattern of kangaroo activity on the runway around last light.

Had the requirement applied, an inspection would have been requested and its result passed before the roll, and the aerodrome's staff disperse animals when they find them. One limitation was recorded. An inspection completed some minutes before departure does not guarantee a clear runway at the moment of the roll. That limitation is why the influence is put at very likely rather than higher. The absence of an inspection on the night is the individual action examined elsewhere. The wildlife plan threshold and the approval of the evening service, which shaped this control, are also examined elsewhere.

The listing criterion for Appendix 8A very likely contributed to the absence of a runway inspection before the first departure after last light.

Rejected take-off training covers aircraft system failures below V1 only

Module 4 of recurrent syllabus TRG-SYL-07 contained four scripted scenarios:

- engine failure at 90 kt
- engine fire at 100 kt
- configuration warning at 85 kt
- engine failure at V1 plus 5 kt, continued.

Each was an aircraft system event. None presented an obstruction, an animal or any other external hazard on the runway. No crew had been given an external obstruction scenario in the five training cycles on file, and 214 crew completed the module in the 2025 to 2026 cycle.

Part B 4.6 matched the syllabus. 4.6.4 states "At or above V1 the take-off shall be continued." The section gives no guidance on an obstruction seen on the runway at any speed. It also gives no amplification for a case in which continuing is itself hazardous. Both flight crew independently described their training scenarios as engine events. It was extremely likely that this gap existed. The small residual doubt reflects that scenario records beyond five cycles were not available.

Whether the gap influenced the decision was argued both ways.

- **For influence.** The first officer's sighting call came at 07:30:31. The reject was initiated at 07:30:32, 6 kt above V1. The captain had not seen the animals himself. The decision was made in about a second, without any prior frame for the question. Preparing crews for that kind of decision is the purpose of the training system.
- **Against influence.** The captain stated that he knew the rule and departed from it deliberately because he judged continuing to be worse. On that reading, obstruction training would not have changed what he knew. The gap would then be a finding about the fleet rather than a cause of this reject.

The investigation accepted influence as likely. An unrehearsed decision made in seconds is not the same decision as one made against a briefed and practised frame. Both readings stand. This finding explains the conditions under which the decision was made. It does not excuse the decision. The decision itself is examined under individual actions.

The absence of any obstruction scenario or guidance in rejected take-off training and procedures likely contributed to the take-off being rejected 6 kt above V1, although the captain's account that he departed from a known rule deliberately supports the alternative reading that it did not.

Cabin floor proximity emergency lighting partially unserviceable

The first officer noticed unlit sections of the aisle floor strip while walking back through the cabin during the disembarkation. A functional check on 24/06/2026, with the system armed and on battery power, found the following:

- Two of the six floor proximity escape path segments, on the left side aft of row 16, did not illuminate.
- The fault was traced to a failed ballast common to both segments.

It was virtually certain that the defect existed.

The defect was undetectable in service. The pre-flight test on this type illuminates the ceiling emergency lights and exit signs. It neither illuminates nor monitors the floor segments. The last dedicated functional test was 11 months before the occurrence, within the maintenance programme's 24-month interval.

The defect was unlikely to have influenced this occurrence. No evacuation was commanded. The main cabin lights were on throughout. Passengers left by the forward airstair without incident, 4 minutes 8 seconds after the aircraft stopped.

It remains a safety issue. A fire indication, smoke, or a night evacuation over the wing would have put passengers on a partly dark aft escape path. Nothing available to the crew would have revealed the defect. The same ballast is fitted across the type. A 24-month interval on a system the daily test cannot see means the fleet's true state is unknown.

The partial failure of the cabin floor proximity emergency lighting, undetectable by any routine check, did not contribute to the occurrence but very likely increased risk to occupants in any evacuation.

Local Conditions

Departure in the hour after last light, the peak of kangaroo movement

The take-off roll began at 17:30 local, 32 minutes after sunset and 5 minutes after the end of civil twilight. The moon was 14% illuminated and did not rise until 22:41. No town lighting was within sight. At a site visit at the same time of day on 23/06/2026, the investigator confirmed that the unlit graded strip either side of the sealed surface could not be seen from the centreline.

The aerodrome's own wildlife sighting log placed the departure inside its period of greatest kangaroo activity. Of 41 kangaroo entries between 01/2025 and 06/2026, 38 were timed between one hour before and two hours after last light. Eleven recorded animals on the sealed surface of runway 09/27. The astronomical data are documentary. The log is a paper register whose completeness could not be verified. It is extremely likely that the departure took place in full darkness within the aerodrome's peak period of kangaroo movement.

The light condition governed what the crew could see. Nothing crossing the strip was visible until it reached the lit surface. The first officer's account on the CVR was that the animals "came out of the dark on the left of the strip". That is a direct consequence of the darkness. The timing also coincided with the logged movement peak.

A departure after last light is routine in itself. Here it was the condition under which a runway inspection before the first departure after last light would have mattered. Why no inspection was requested is argued under Risk Controls.

The departure in darkness during the peak of kangaroo movement likely contributed to two kangaroos entering the runway strip and moving onto the sealed surface undetected.

Runway edge lighting intensity reduced the crew's ability to detect the animals

The investigation examined whether the runway edge lighting impaired detection of the animals. The matter could not be resolved.

Evidence supporting the proposition came from a photometric check carried out at the investigator's request on 25/06/2026. Four northern edge lights between the 900 m and 1,200 m markers measured 118% to 131% of nominal intensity for stage 3. The remaining lights were within tolerance. The out-of-tolerance lights were in the section the animals crossed. The aerodrome operator's maintenance programme checked lamp serviceability but did not measure intensity. In interview, the captain described the lighting as brighter than he would want. He described a corridor of light with everything outside it black, and said anything off the seal could not be seen coming.

Evidence against it came first from the first officer, who was the crew member who saw the animals. She described the lighting as normal. She said she saw the animals clearly once they were on the sealed surface, at around 200 m. She also said that, with only two daylight sectors into Tarlton Springs to compare against, she was not well placed to judge whether the lighting was unusual. The captain's non-detection does not depend on the lighting. He stated on the CVR, "I was heads-in on the engine page after V one", which is where the procedure placed him.

The deviation was real and in the relevant section of runway. However, the check was made seven days after the occurrence. No measurement exists of the lights on the night. Nothing establishes a link between an edge light at 131% of nominal and the detection of an object on the runway. The two crew accounts are both credible and do not agree. Resolving the matter would require a measurement of the lighting as it was on the night, and evidence linking that level of intensity to detection performance. Neither is available.

It is about as likely as not that the lighting reduced the crew's ability to detect the animals. The matter remains unresolved and forms no part of the findings.

Take-off commenced with a tailwind above limits

A tailwind was ruled out, because two independent sources gave a headwind throughout the roll. The 06:30, 07:00 and 08:00 UTC observations gave a component of 5 to 7 kt, and the recorded FDR wind gave 4 to 6 kt. Both were consistent with the 5 kt headwind used in the crew's performance entry. It is very unlikely that any tailwind existed.

Fifth sector, 11 hours into duty, 40 minutes behind schedule

It is virtually certain that the crew signed on at 06:30 local for a five-sector duty. The fifth sector was added that morning when the rostered crew reported unfit. The occurrence was 11 hours 00 minutes into a duty planned at 12 hours 05 minutes, against a limit of 12 hours 30 minutes. The aircraft departed 40 minutes behind schedule. Both crew described themselves in interview as tired but not struggling.

A late fifth sector at that point in a duty is where degraded decision making would be expected. The recorded data did not show it. Power levers reached flight idle within 0.6 seconds of the reject decision. Maximum braking was established 2 seconds after the decision, against a fleet mean of

3.2 seconds. No step was omitted during the roll. The one decision the duty could plausibly have affected was whether to request a runway inspection. No document prompted either crew member to consider that step, so fatigue could not have caused its omission. It is unlikely that the duty influenced the occurrence.

The exposure remains. The evening Tarlton Springs rotation routinely places the last sector in the last hour of a 12-hour duty. It operates to an aerodrome with no company engineering and no alternate within 200 nm at night. It also has no crewing resilience, so a single unfit crew member turns a four-sector day into a five-sector day on the morning of operation.

The scheduling of the evening Tarlton Springs rotation likely increased risk to operations, although it did not contribute to this occurrence.

Individual Actions

Take-off rejected 6 kt above V1

V1 was called at 112 kt at 07:30:29. The first officer's sighting call followed at 07:30:31. The captain's response came at 07:30:32: "Stopping. I have control." Both power levers were at flight idle within 0.6 seconds, at 118 kt, three seconds and six knots past V1. The recorded data, the recorded audio and the captain's own account agree. It is virtually certain that the take-off was rejected above V1.

Part B 4.6.4 of the operator's manual states that "At or above V1 the take-off shall be continued." Section 4.6 contains no guidance on an obstruction seen on the runway, at any speed. It also gives no amplification of 4.6.4 for a case in which continuing is itself hazardous. The captain stated in interview and in his air safety report that he weighed the options this way:

- rotating into two large animals would leave the aircraft airborne at night with damage that could not be assessed, over country with no usable option; and
- rejecting was a risk whose end he could see.

He acknowledged that the decision was not in accordance with the procedure. The decision rested on the first officer's call alone. The captain later said on the CVR: "I never had them. I was heads-in on the engine page after V one."

The scheduled accelerate-stop distance of 1,780 m assumes the reject begins at or before V1. The margin between the accelerate-stop distance required and the distance available was 67 m. At the reject the aircraft was 1,152 m along runway 09, with 695 m of sealed surface remaining. The stop actually achieved from 118 kt took 757 m. Beginning the stop six knots and three seconds late consumed the 67 m margin and more, before the loss of No. 2 reverse is counted. The shortfall in deceleration after the strike is a separate factor, set out under Occurrence Events.

The decision was taken in the conditions it was taken in. The animals were on the runway because of the conditions set out under Local Conditions, Organisational Influences and External Influences. The operator's rejected take-off training addressed aircraft system failures below V1 only, and that finding is argued under Risk Controls. The matter of runway edge lighting intensity is unresolved and is discussed under Local Conditions. Nothing from it is carried here.

It is extremely likely that rejecting the take-off above V1 contributed to the aircraft being unable to stop on the sealed surface.

No runway inspection requested before the first departure after last light

Neither the CVR nor the tower's recorded audio contains a request for a runway inspection, an offer of one, or any mention of wildlife before departure. No wildlife transmission appears on the tower channel in the preceding 90 minutes.

The aerodrome's daily inspection log for 18/06/2026 records one inspection, at 07:35 local, before the first arrival. It recorded the movement area as clear. The log records no later inspection and no request for one. It is virtually certain that no inspection was requested before SB512 departed.

The facility existed. Aerodrome Manual 6.3 provides for inspections outside the scheduled time at the request of the reporting officer or a pilot in command. The reporting officer confirmed that an inspection would have been carried out if asked.

Neither crew member considered it, and no document asked them to. Part A 8.14.2 requires an inspection before the first departure after last light only at aerodromes listed in Appendix 8A. Part A 8.14.3 requires no specific action at any other aerodrome. Tarlton Springs was not listed and had never been listed. The captain had expected it to be on the list and was surprised to find it was not. The listing criterion is argued under Risk Controls.

The aerodrome's own wildlife sighting log held relevant knowledge that did not reach the crew. It recorded 41 kangaroo entries between 01/2025 and 06/2026:

- 38 of those entries were timed between one hour before and two hours after last light; and
- eleven recorded animals on the sealed surface of runway 09/27.

The log is a paper register held in the aerodrome office. The reporting officer confirmed that no airline had ever asked for it.

This was an omission with no owner. Nothing in the operator's documents directed either crew member to consider an inspection, and the aerodrome would have carried one out on request.

An inspection would have covered the section between 1,050 m and 1,250 m. The fence pad and both carcasses were found in that section, and aerodrome staff disperse animals when they find them. However, an inspection some minutes before the roll cannot guarantee the runway stays clear.

It is likely that the absence of a runway inspection contributed to the animals being on the sealed surface during the take-off roll.

Maximum braking and No. 1 reverse applied without delay

The following sequence is established by the recorded data:

- The power levers reached flight idle 0.6 seconds after the decision.
- Maximum braking was established at 07:30:34, 2 seconds after the decision. The operator's recorded fleet mean is 3.2 seconds, which places this response at the fast end of the fleet distribution.
- Anti-skid was active on all four wheels.
- Brake energy reached 84% of the certified maximum without a brake fire.
- No. 1 reverse was selected at 07:30:36. No. 2 reverse was unavailable because its propeller had been damaged in the strike.
- The first officer made the speed callouts and the runway-end call.

The aircraft left the seal at 21 kt and stopped 62 m into the 90 m RESA. It came to rest upright, on the extended centreline. Passengers were disembarked by the forward airstair without an evacuation.

A stop begun at the fleet mean response would have carried the aircraft an estimated further 30 to 40 m. The rejected take-off was outside procedure, but the stop that followed was executed faster than the fleet average.

It is very likely that the prompt application of maximum braking and No. 1 reverse reduced the severity of the overrun.

Occurrence Events

Two kangaroos entered the runway strip and moved onto the sealed surface

The animals' presence and final positions are not in doubt. Two adult kangaroos were recovered from runway 09 at 1,188 m and 1,221 m from the threshold, on the northern half. The first officer called them at 07:30:31: "Roos - roos, roos, on the left, they're on the -". Shortly after the aircraft stopped, the first officer described them on the flight deck audio: "They came out of the dark on the left of the strip, crossed the centreline." The first officer's estimated range at first sight was around 200 m.

How the animals reached the runway is less certain. Since 04/2023 the northern perimeter fence had been recorded as failed, with animals observed passing through. On inspection after the occurrence it was a continuous 140 m gap, with an animal pad through it at the 1,050 m mark. That pad is on the same side of the runway as the carcasses and 138 m short of the first one. The physical evidence best supports the pad as the entry point. However, the fence was failed over 140 m, and the exact crossing point could not be established.

The event was not unusual for the aerodrome. The sighting log held eleven entries recording kangaroos on the sealed surface of runway 09/27, all around last light. No runway inspection had been carried out since 07:35 local that morning. The fence deferral, the timing of the departure and the absence of an inspection are analysed in their own lanes.

The reject and the strike both follow directly from the animals being on the sealed surface. Whether the runway edge lighting affected the crew's ability to detect the animals is discussed separately as an unresolved matter.

It was very likely that two kangaroos entered the runway strip and moved onto the sealed surface, and extremely likely that their presence there led to both the rejected take-off and the strike.

No. 2 propeller and right main gear struck two kangaroos

The strike occurred at 07:30:33, at 114 kt, one second after the reject was initiated. Several sources agree on it:

- The FDR recorded a lateral acceleration spike of 0.31 g right and a vertical spike of 1.48 g.
- The CVR recorded a sound consistent with a heavy impact, followed by the first officer's call that the aircraft had "took one on the prop."
- Three No. 2 propeller blades were damaged beyond repairable limits, one with a separated tip, and animal tissue was recovered from two blade roots.
- The right main gear doors were distorted, and propeller debris had punctured the lower right fuselage skin.

The strike's consequence for the stop is what matters here. No. 2 propeller RPM decayed from the moment of impact, and the No. 2 reverse interlock did not make. When reverse was selected at 07:30:36, it was available on No. 1 only. The captain's call at that moment was "Reverse - number two's not coming." The scheduled accelerate-stop figures assume symmetric reverse. From about one second into the stop, the aircraft no longer had it.

It was virtually certain that the No. 2 propeller and right main gear struck both kangaroos, and virtually certain that the strike, by removing No. 2 reverse, reduced the deceleration available for the rest of the stop.

Deceleration achieved below the scheduled accelerate-stop performance

The scheduled accelerate-stop distance required was 1,780 m against an ASDA of 1,847 m, a margin of 67 m. That figure rested on two assumptions: that the stop began at or before V1, and that symmetric reverse was available. Neither held.

The first departure was the late start. Distance reconstruction from recorded groundspeed places the aircraft 1,152 m along runway 09 when the take-off was rejected at 118 kt, leaving 695 m of sealed surface. The stopping distance achieved from that speed was 757 m. The 62 m difference matches the aircraft's resting position in the RESA. The physical marks agree with the reconstruction: continuous tyre marks run from 1,160 m to the runway end, and the nose wheel rut ends 62 m beyond it on the extended centreline.

The second departure was the reduced deceleration. The mean deceleration under maximum braking was 2.6 m/s², approximately 15% below the value used in the certified data for this weight and runway condition. The braking itself delivered what was asked of it, so the shortfall does not lie in the braking system:

- Maximum braking was established two seconds after the decision.
- Anti-skid was active on all four wheels and tested serviceable afterwards.
- Brake energy absorbed was 84% of the certified maximum, with no brake fire or hydraulic leak.

The difference from the scheduled figure corresponds to the loss of No. 2 reverse. Once the aircraft left the seal at 21 kt, deceleration in the graded gravel fell to 1.9 m/s². The aircraft stopped 62 m into the 90 m RESA, 28 m short of its end.

The late start and the reduced deceleration are not independent. The investigation assessed that both were needed to carry the aircraft beyond the runway end and that neither alone would have done so. The overrun is therefore treated as a single mechanism: a stop begun 6 kt above V1 with a 67 m scheduled margin, and then deprived of symmetric reverse.

It was virtually certain that the deceleration achieved fell below the scheduled accelerate-stop performance, and virtually certain that this shortfall, combined with the late start, resulted in the overrun onto the runway end safety area.

No. 2 engine power loss during the take-off roll

A No. 2 engine power loss before the reject was examined and ruled out as very unlikely, for four reasons:

- Both engines held take-off torque within 0.1% of each other until the power levers were retarded at 07:30:32, with no caution, warning or exceedance.
- The CVR contains no engine-related call.
- The borescope of both engines found no pre-existing damage.
- The No. 2 RPM decay began at 07:30:33, at the moment of impact and after the reject.

Conclusions

These findings are made with respect to the occurrence and should not be read as apportioning blame or liability to any organisation or individual.

Findings

Contributing safety factors

1. It was virtually certain that Aerodrome Standard AS-139 10.4.2 did not require Tarlton Springs to hold a documented wildlife hazard management plan, because the aerodrome fell below both size limbs of the standard and was not in breach. The absence of a plan likely contributed by leaving the aerodrome's wildlife measures without an owner, an audit trail or a funded programme. This link was less firmly established than the other contributing factors, because a documented plan would not necessarily have secured funding for the fence repair. (Safety issue)
2. The aerodrome operator deferred repair of the failed northern perimeter fence at three successive budget rounds, and this was virtually certain. The fence had been recorded in 04/2023 with animals observed passing through it. The deferral very likely contributed by leaving a 140 m breach through which kangaroos could reach the movement area. (Safety issue)
3. Two kangaroos very likely entered the runway strip and moved onto the sealed surface of runway 09 ahead of the aircraft. The animal pad through the failed fence at the 1,050 m mark was the entry point best supported by the physical evidence, but the exact crossing point was not established. The animals' presence on the runway very likely contributed to both the rejected take-off and the strike.
4. With virtual certainty, the pilot in command rejected the take-off at 118 kt, 3 seconds and 6 kt above V1, outside the operator's procedure to continue at or above V1. This very likely contributed to the overrun, because the scheduled accelerate-stop distance of 1,780 m assumed a stop begun at or before V1, and only 695 m of sealed surface remained when the stop began.
5. The deceleration achieved was below the scheduled accelerate-stop performance, and this was virtually certain. The mean deceleration was 2.6 m/s², approximately 15% below the certified value. With virtual certainty, this contributed to the aircraft leaving the sealed surface at approximately 21 kt and stopping 62 m into the runway end safety area.
6. The departure very likely took place in the hour after last light, 5 minutes after the end of civil twilight and with no moon, which was the period in which the aerodrome's own log showed almost all kangaroo movement. This likely contributed, because nothing off the lit sealed surface could be seen and the animals were not detected until they were on it.
7. It was virtually certain that the operator's Operations Manual Part A 8.14 required a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A. The only criterion for listing an aerodrome was a reported wildlife strike involving a company aircraft, so Tarlton Springs could not be listed on its sighting history. This very likely contributed to no inspection being required before the departure. (Safety issue)
8. No runway inspection was requested before the first departure after last light, and this was virtually certain. No document prompted either crew member to consider one, although the aerodrome would have carried it out if asked. The absence of an inspection likely contributed to the animals being on the runway at the roll, although an inspection completed some minutes before departure would not have guaranteed a clear runway.

9. The operator very likely approved the evening service as a schedule change without reopening route risk assessment RRA-041. RRA-041 restricted Tarlton Springs to daylight operations. Not reopening it likely contributed to the operation proceeding without its wildlife exposure being assessed, although a review might have addressed lighting and fuel alone. (Safety issue)

10. The operator's rejected take-off training very likely covered only aircraft system failures, three of them below V1, and gave no guidance on an obstruction on the runway at any speed. This likely contributed to the decision to reject after V1 being made without any prior frame. However, the pilot in command stated that he knew the procedure and departed from it deliberately, so the training gap may bear on the fleet more than on this decision. (Safety issue)

11. It was virtually certain that the No. 2 propeller and right main gear struck both kangaroos at 114 kt, one second after the reject began. With virtual certainty, the strike contributed to the overrun because it disabled the No. 2 reverse and left reverse thrust available on No. 1 only.

Other factors that increased risk

12. With virtual certainty, two of the six cabin floor proximity escape path segments aft of row 16 on the left side were dark because of a failed common ballast. The defect could not be detected by the pre-flight test, and the functional test interval is 24 months. It did not influence this occurrence, but it very likely increased risk to occupants in any evacuation on the fleet, because the same ballast is fitted across the type. (Safety issue)

13. It was virtually certain that the occurrence sector was the fifth of the day, flown 11 hours into the duty and 40 minutes behind schedule, after being added that morning when the rostered crew reported unfit. The recorded crew response times did not support it having influenced this occurrence. The exposure is built into the evening schedule and likely increased risk on that service.

Other findings

14. It was very unlikely that the take-off commenced with a tailwind. The aerodrome observations and the recorded data both showed a headwind component throughout the roll, and this hypothesis was ruled out.

15. It was very unlikely that the No. 2 engine lost power during the take-off roll. Both engines were matched at take-off torque until the reject, and the No. 2 propeller RPM decay began at the moment of impact, so this hypothesis was ruled out.

16. With virtual certainty, maximum braking was established 2 seconds after the decision against a fleet mean of 3.2 seconds, and No. 1 reverse was applied without delay. This very likely reduced the severity of the outcome, because a stop begun at the fleet mean response would have carried the aircraft an estimated further 30 to 40 m, close to the far edge of the runway end safety area.

The investigation could not determine whether the runway edge lighting intensity reduced the crew's ability to detect the animals, and it was about as likely as not that such an effect existed.

- **Evidence for an effect.** A check seven days after the occurrence measured four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal for stage 3. This was in the section the animals crossed. The pilot in command also described the lighting as a corridor of light with darkness either side.
- **Evidence against an effect.** The first officer saw the animals and described the lighting as normal, although she had only daylight landings there to compare against. The pilot in command's non-detection is explained by his attention being on the engine page after V1, as the procedure required.

- **Why it could not be resolved.** No measurement exists of the lighting on the night. Nothing establishes a link between the measured intensity and the detection of an object on the runway, and the matter could not be resolved with the evidence available.

Probable cause

The take-off was rejected 6 kt above V1 when two kangaroos moved onto the runway after last light, and the No. 2 propeller then struck the animals and left reverse thrust available on No. 1 only. Together, these reduced the deceleration below the scheduled accelerate-stop performance, which resulted in the aircraft overrunning 62 m into the runway end safety area.

Contributing factors

- The aerodrome operator deferred repair of the failed northern perimeter fence for three budget cycles, which very likely left a route for kangaroos onto the movement area.
- AS-139 did not require Tarlton Springs to hold a documented wildlife hazard management plan, which likely left its wildlife measures unowned and unfunded. This was the least firmly established contributing factor.
- The departure took place in the hour after last light, when kangaroo movement peaked and nothing off the sealed surface could be seen, which likely contributed.
- The operator required a runway inspection only at aerodromes listed after a reported strike, which very likely meant none was required at Tarlton Springs.
- No runway inspection was requested before the departure, which likely contributed to the animals being on the runway at the roll.
- The evening service was approved without reopening the daylight-only route risk assessment, which likely contributed.
- The operator's rejected take-off training did not address runway obstructions, which likely contributed to the decision to reject after V1, although the pilot in command stated he departed from the procedure knowingly.

Recommendations

Documented wildlife plan required only above a size threshold the aerodrome does not meet

Aerodrome Standard AS-139 section 10.4.2 required a documented wildlife hazard management plan only at aerodromes that met one of two conditions:

- used by aeroplanes above 27,000 kg MTOW, or
- serving more than 350,000 passenger movements a year.

Tarlton Springs met neither. Its heaviest scheduled aeroplane was 22,900 kg, and it handled 61,400 passenger movements in 2025. Under section 10.4.3, its wildlife measures did not need to be documented at all. The aerodrome was not in breach.

The significance of the issue lies in what the threshold left in place. The aerodrome's wildlife programme was undocumented, unaudited and, in practice, unfunded.

- **Sighting log:** The aerodrome's own log held 41 kangaroo entries, 38 of them around last light and eleven on the sealed surface. No operator had ever asked to see it.
- **Fence repair:** The northern fence was recorded as failed in 04/2023 and deferred for three budget cycles.
- **Runway inspection:** The criterion for listing an aerodrome for runway inspection could only be met by a reported strike.

It was likely that the threshold contributed to the occurrence, through the fence repair deferral and the runway inspection listing criterion. This conclusion is inferred and less certain than the other contributing factors. Section 10.4.1 still obliged the aerodrome operator to take reasonable measures. There was no evidence that a documented plan would have been funded any better.

No safety action has been recorded against this issue, and no recommendation is made.

Perimeter fence repair deferred for three budget cycles

The aerodrome operator deferred repair of the northern perimeter fence at Tarlton Springs through three consecutive budget rounds. In 04/2023 the operator's works register recorded four displaced panels and strained wire between the 700 m and 1,300 m runway markers, with animals observed passing through, and estimated replacement at \$41,000. The item was deferred at the 2023, 2024 and 2025 budget rounds, each time as 'defer, unfunded, review next cycle'. A later entry in 02/2026 recorded that the same section had deteriorated further. On 23/06/2026 the investigation found a continuous 140 m section with the bottom two wires separated from the posts and a clear animal pad through the fence at the 1,050 m mark. It is virtually certain that this condition existed.

The issue matters because the fence, in the condition found, did not exclude kangaroos. The animal pad through the breach lay on the same side of the runway as the strike position, upstream of where the carcasses were recovered. The deferral very likely contributed to two kangaroos entering the runway strip and moving onto the sealed surface. The operator's own register had recorded the breach, and animals passing through it, three years before the occurrence. (Safety issue)

No safety action has been recorded against this issue, by the aerodrome operator or any other organisation. No recommendation is made.

Evening service approved as a schedule change, route risk assessment not reopened

The operator's change management process approved a scheduled evening service into Tarlton Springs as a schedule change, without reopening the current route risk assessment. That assessment, RRA-041 issue 3, restricted operations at Tarlton Springs to daylight. It is extremely likely that the change was approved in this way. It is likely that this contributed to the occurrence.

The change put a company aircraft into Tarlton Springs after last light for the first time. It did so against a restriction that was still in force in the operator's own current documentation. The process asked whether the change was new. It did not ask whether the existing assessment still held.

A review of RRA-041 against a night operation would have reached two matters in the same pass:

- the aerodrome's wildlife exposure
- the Operations Manual Part A Appendix 8A listing, which governs whether a runway inspection is required.

The contribution is assessed as likely rather than certain. Such a review might have addressed night lighting and diversion fuel alone, because those are the matters the restriction names.

No safety action has been recorded against this issue, and no recommendation is made.

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

The operator's Operations Manual Part A 8.14.2 requires the pilot in command to request a runway inspection before the first departure after last light at any aerodrome listed in Appendix 8A, and not to commence the roll until the result is passed. Part A 8.14.3 requires nothing at other aerodromes. An aerodrome is listed only where a wildlife strike involving a company aircraft has been reported there in the preceding 24 months. No provision admits an aerodrome on any other evidence, and no provision asks a crew to report a sighting. Nine aerodromes were listed. Tarlton Springs never had been. The list was last reviewed on 03/03/2025, four months before the evening service began. The existence of this limitation was virtually certain.

The requirement was the operator's own defence against this kind of occurrence, and it was unarmed. The aerodrome's wildlife sighting log held eleven entries of kangaroos on the sealed surface. None of these could place the aerodrome on the list, because they were not strikes and were not reported to the company. The trigger could only be armed by the event the control existed to prevent. Had the requirement applied, an inspection would have been requested and a pass required before the roll, and the aerodrome's staff disperse animals when they find them. The limitation very likely contributed to the absence of a runway inspection before the occurrence departure.

No safety action has been recorded against this safety issue. [NOT ESTABLISHED: safety action taken or recommendation issued to the operator in respect of the Appendix 8A listing criterion]

Rejected take-off training covers aircraft system failures below V1 only

The operator's rejected take-off training very likely prepared crews only for aircraft system failures, almost all of them below V1. All four scripted scenarios in module 4 of TRG-SYL-07 were aircraft system events:

- engine failure at 90 kt
- engine fire at 100 kt
- configuration warning at 85 kt
- engine failure at V1 plus 5 kt, with the take-off continued

No crew had been given an external obstruction scenario in the five training cycles on file. The current cycle alone covered 214 crew.

Operations Manual Part B section 4.6 matched the syllabus. It listed reject items to V1, stated that the take-off shall be continued at or above V1, and gave no guidance on an obstruction on the runway at any speed. Both pilots independently said that every stop-or-go scenario they had been given was an engine event arriving below V1.

It is likely that this gap contributed to the take-off being rejected 6 kt above V1. The captain made that decision within about two seconds, with no rehearsed approach to the situation. However, the captain said he knew the rule and departed from it deliberately because he judged continuing to be worse. On that reading, training would not have changed what he knew. Both readings are held. The gap explains the conditions of the decision; it does not excuse it.

The issue applies across the operator's fleet, not only to this crew.

No safety action has been recorded against this issue.

Cabin floor proximity emergency lighting partially unserviceable

Two of the six floor proximity escape path segments aft of row 16 on the left side of VH-SBQ were dark on battery power with the system armed. A functional check on 24/06/2026 traced the fault to a failed common ballast. The first officer had noticed sections of the aisle floor strip unlit while walking back through the cabin during the disembarkation.

The defect was unlikely to have influenced the overrun or its outcome. No evacuation was commanded and the main cabin lights were on throughout. Passengers disembarked by the forward airstair in 4 minutes 8 seconds without incident.

The defect very likely increased risk to occupants in any evacuation. The pre-flight emergency lighting test on the type illuminates the ceiling lights and exit signs, but it neither illuminates nor monitors the floor segments. Neither the crew nor a line engineer could detect the defect without a dedicated functional test. The last such test was 11 months before the occurrence, within the 24-month programme interval. That interval applies to a system the daily test cannot see, so the true state of the fleet is unknown, and the same ballast is fitted across the type. A fire indication, smoke or a night evacuation over the wing would each change the situation. In any of those cases, two dark segments in the aft cabin could turn an orderly exit into a blocked one.

No safety action has been recorded against this issue.

Distribution

Title	Name
Head of Flying Operations	R. Calvert
Head of Safety	M. Falkiner
Fleet Manager, Turboprop	D. Osei
Training Manager	S. Arnott
Aerodrome Reporting Officer, Tarlton Springs	K. Pryor